THERMAL EQUIPMENT

Let's Combat Illegal Garbage Dumping!
When mixed together, waste becomes mere garbage; however, when separated, it transforms into valuable resources.

We provide a diverse array of equipment, including compact units for residential waste and large-scale systems designed for industrial waste, along with specialized thermal equipment.

Cleaner Communities. Sustainable Recycling Solutions.

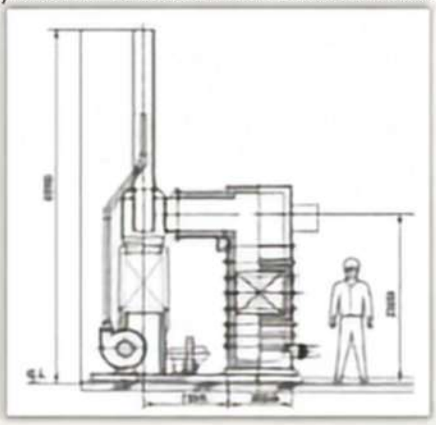
THE PRYMM GROUP
Exclusive Distributor for the Philippines

THERMAL EQUIPMENT (KGV-20)

MODEL: **KGV-20 WG**

Processing Power	1,000kg/ day
Furnace Size	25.06 X 806 mm
Size of Input Slot	400 X 600
Floor Area of the Furnace	3.23 ft2
Volume of the Primary Heat Treatment	chamber 0.44 m3
Volume of the Secondary Heat Treatment	chamber 0.23 m3
Output Secondary Combustion Burner	0.2kw
*Fuel Consumption of Secondary Burner	8L/h
Chimney Height and Diameter	5,000mm X 250 mm
Fuel Tank White Kerosene	187L
Electric Blower 3-Phase	2

- The secondary burner is used when black smoke is emitted.



Efficient & Eco-Friendly Waste Solution

Our advanced thermal equipment is designed to handle a wide range of waste materials, including combustible waste from composting facilities, household waste, and medical waste such as strings and rubber gloves. It also efficiently processes wood chips and waste plastics. Built as a compact unit mounted on a durable 72 ft² steel plate, this system offers both space efficiency and robust performance.

No main fuel is needed; once you light the pilot, the airflow keeps it burning steadily. A secondary burner is attached to the combustion chamber's exhaust outlet; if black toxic gas is emitted, activate the secondary burner. It burns off the toxic gas.

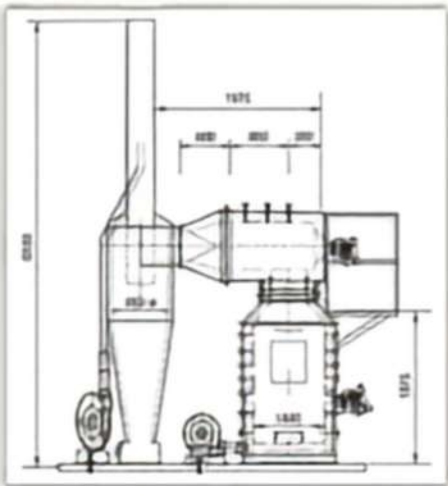
Operational Costs:
* Electricity consumption is limited to the air blower fan operation
* Kerosene is only used when the secondary burner is activated (rarely required during normal operatio

Even if non-combustible materials such as stones, cans, or glass fragments are accidentally mixed in, they ca
recovered together with the ash.

Waste volume is reduced to as little as 1/100, producing non-toxic residual ash.

THERMAL EQUIPMENT (KGV-45)

Recommended for composting facilities, seafood markets, food processing plants, hospitals, waste collection sites, supermarkets and farms.



MODEL: KGV-45 WG

Processing Power 100kg/h	2000kg/day
Furnace Size Height/Width	336OX 162OX 1310
Size of Input Slot	600 X 600
Floor Area of the furnace	861 ft2
Volume of the Primary Heat Treatment	chamber 130 m3
Volume of the Secondary Heat Treatment	chamber 0.73 m3
Output Secondary Combustion Burner	0.25kw
+Fuel Consumption of Secondary Burner	15L/1h
Chimney Height and Diameter	6.000mm X 400mm
Fuel Tank White Kerosene	187L
Electric Blower 3-Phase	2

- *The secondary burner is used when black smoke is emitted.*

High-Temperature Waste Treatment

This system is designed for the thermal processing of a diverse array of materials, such as:

- Combustible waste from composting facilities
- Food waste from restaurants and households
- Industrial and medical waste
- Wood chips
- Plastics

It operates at temperatures exceeding 800°C for effective and thorough reduction, without the need for primary fuel. To ignite the system, an oil-soaked cloth is placed at the base, followed by layers of paper, plastics, wood chips, and rubber materials. As the airflow increases, resulting in intensified combustion, food and medical waste can be introduced to enhance processing efficiency. In instances of overload that may lead to incomplete combustion, restoring performance is simple: just boost the airflow and activate the secondary burner system. Operators can easily monitor combustion conditions through the viewing window and gradually add waste to ensure optimal burning performance.

Animal bones, such as those from pigs or poultry, are discharged in ash form while retaining their original shape; however, they are fully carbonized and can be easily crushed. Non-combustible materials such as glass and concrete fragments will not burn and can be collected along with the ash after processing.

Warning: Chlorine-based materials, including PVC pipes and decorative laminates, must not be combusted, as they may release harmful substances and cause damage to the furnace.

THERMAL EQUIPMENT (KGV-75)

Built for efficient treatment of long-exposed waste piles, this high-performance system delivers reliable and powerful operation in demanding environments. A large 2.0 m × 0.95 m feed opening allows easy loading with backhoes or power shovels, reducing manual work. The water-cooled, double-door inlet enhances safety and heat control.

The primary combustion chamber features durable refractory lining and external water cooling, with a high-efficiency burner and fan system to maximize combustion. Secondary and tertiary chambers, also burner-equipped, ensure complete incineration, while the tertiary chamber is supported by a water-cooling pump. An advanced exhaust system with an induced draft fan efficiently manages heat and airflow across all combustion stages, supporting stable operation and cleaner emissions. Engineered for high-capacity, efficient waste processing—combining performance, safety, and environmental responsibility in a compact design.

MODEL: **KGV 75**

Direct-feed Type

Processing Power 100kg/h	*5,000kg/day*	
Furnace Size Height/Width	*2240X 52400X 2275*	
Size of Input Slot	*950 X 1900*	*Water-cooled, Wdoors*
Floor Area of the Furnace	*19.36 ft2*	
Volume of the Primary Hear Treatment	*chamber 3.30 m3*	*Water-cooled, wall*
Volume of the Secondary Heat Treatment	*chamber 3.26 m3*	
Output Secondary Combustion Burner	*0.4kw*	
**Fuel Consumption of Secondary Burner*	*15L/h*	
Chimney Height and Diameter	*8,990mm X 600mm*	
Fuel Tank White Kerosene	*187L*	
Electric Blower 3-Phase 200v	*3*	*(Equipped with a smoke extraction fan)*
The secondary burner is used when black smoke is emitted.		*The walls inside the furnace are made of fire-resistant material.*
Power Consumption	*21 kw/h*	
Water:Cool To Combustion Chamber	*150L/h*	
White Kerosene	*15L/h*	

THERMAL EQUIPMENT (KGH 600)

Production days/270 **Assembly days/45**
Processing Power 600kg/h **13,200kg/day Site Area 1,399 sq.ft.**

Engineered for disposing of large industrial waste, as well as trees from farms and large volumes of household waste.

Site Area: 16,000mm (width) X 8,100 mm (height) = 1,399 sq.ft.

The spacious opening allows for effortless loading, even of lengthy items. The furnace reaches temperatures exceeding 1,000 degrees, making it suitable for disposing of items like motorcycle tires. Fuel is combusted in both the primary and secondary combustion chambers, while water circulates from the tank to various parts of the furnace, ensuring the entire unit remains cool. Additionally, the hot water can be utilized for heating a swimming pool before it's released.

MODEL: KGV 600 Direct-feed Type

Processing Power 600kg/h	*12,000kg/day*	*20h*
Size of Input Slot	*W 3,160 X D 1,680*	*Water-cooled, W-door*
Primary Combustion Reactor	*L3080X W1700X D1680*	*Water-cooled walls & Fire-resistant walls*
Secondary Combustion Chamber Hear Treatment	*L2000X W1890X D2230*	*Combustion Burner & Fan*
Volume of the Secondary Heat Treatment	*chamber 0.73 m3*	
Output Secondary Combustion Burner	*0.25kw*	
**Fuel Consumption of Secondary Burner*	*5-20L*	
Chimney Height and Diameter	*H 10,000 X 900*	
Fuel Tank White Kerosene	*187L*	
Electric Blower 3-Phase 200v	*3*	*(Equipped with a smoke extraction fan)*
The secondary burner is used when black smoke is emitted.		
Power Consumption	*78 kw/h*	*FOR BLOWER*
Water will be used in the section (Inlet, primary combustion chamber, cooling pump, cooling tower)	*3600L/h*	

THE THERMAL CYCLE: TRANSFORMING WASTE INTO RESOURCES

WASTE FEEDING & INPUT

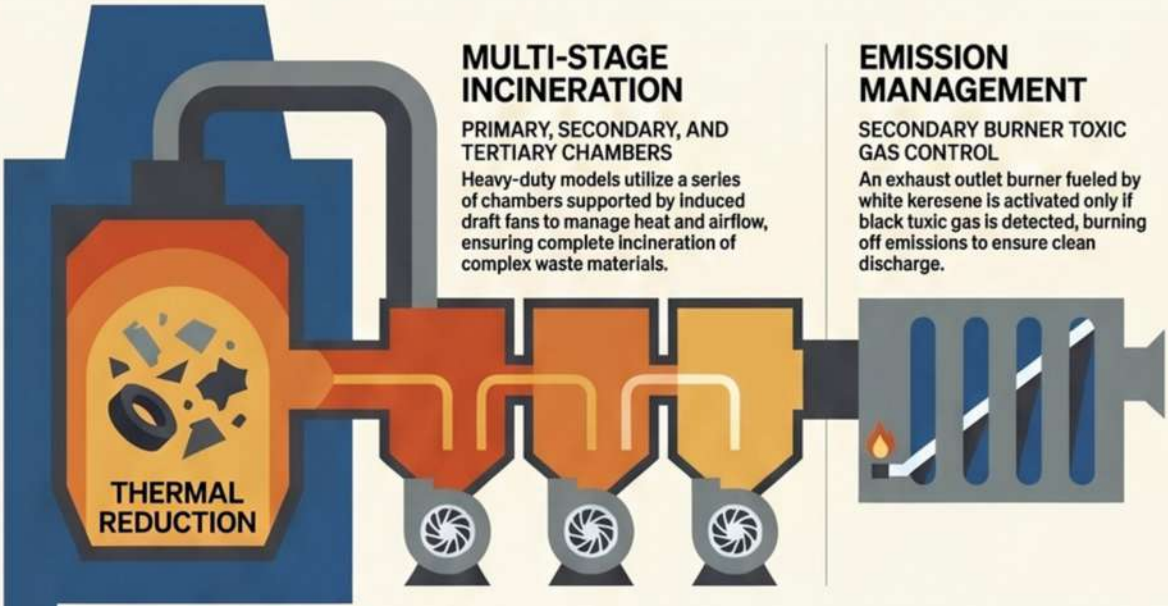
HIGH-CAPACITY LOADING VIA DIRECT-FEED

Large-scale units like the KGV 75 feature 2.0m x 0.85m openings for loading via backhoes, while compact units use specific input slots (up to 600x600mm) for manual feeding.

AIRFLOW-DRIVEN IGNITION

PILOT-START WITH ZERO PRIMARY FUEL

The system is initiated with a pilot (often an oil-soaked clufh and paper); once lit, electric blowers provide forced airflow to intensity and sustain combustion without a continuous main fool source.



MULTI-STAGE INCINERATION

PRIMARY, SECONDARY, AND TERTIARY CHAMBERS

Heavy-duty models utilize a series of chambers supported by induced draft fans to manage heat and airflow, ensuring complete incineration of complex waste materials.

EMISSION MANAGEMENT

SECONDARY BURNER TOXIC GAS CONTROL

An exhaust outlet burner fueled by white keresene is activated only if black tuxic gas is detected, burning off emissions to ensure clean discharge.

HIGH-TEMPERATURE CORE COMBUSTION

Materials are processed at temperatures exceeding 800°C (and cip 1,000°C in industrial models), ensuring thorough reduction of waste, including durable items like motorbike tires.

SYSTEM COOLING

INTEGRATED WATER-COOLING INFRASTRUCTURE

High-capacity units circulate water (up to 3,800L/h) through furnace walls, doors, and cooling towers to maintain safe operational temperatures.

RESOURCE RECOVERY

ASH COLLECTION AND MATERIAL SALVAGE

Final output consists of ash and fully carbonized organic matter (like bones); non-combustibles such as glass, stones, and concrete fragments remain intact and are safely recovered.

ENERGY REPURPOSING

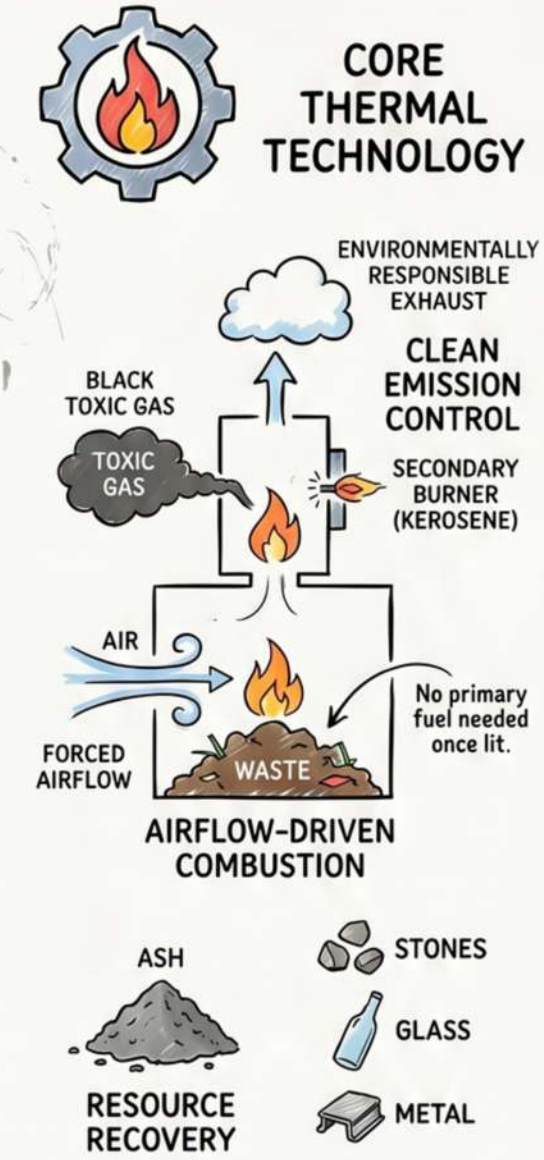
SECONDARY HEAT UTILIZATION

Hot water generated by the cooling system can be captured before discharge and repurposed for facilities like heated swimming pools.

MODEL	TYPE	PROCESSING POWER (HOURLY)	MAX DAILY YIELD
KGV-20 WG	Compact	**50** kg/h	**1,000** kg/day
KGV-20 WG (HT)	High-Temp	**100** kg/h	**2,000** kg/day
KGV 75	Heavy-Duty	**100 – 200** kg/h	**5,000** kg/day
KGV 600	Industrial	**600** kg/h	**13,200** kg/day

Eco-Efficient Thermal Waste Solutions: Transforming Garbage into Resources

PHILOSOPHY: Mixed together = garbage. Separated = resources.

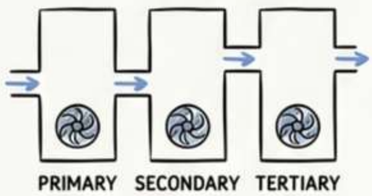


EQUIPMENT LINEUP COMPARISON

KGV-20 WG (Compact Unit)	KGV-20 WG (High-Tomperatore Unit)	KGV 75 (Heavy-Duty Direct-Feed)	KGV 600 (Industrial Direct-Feed)
50kg/h (1,000kg/day)	800°C+		1,000°C+
Ideal for: household waste, medical syringes, rubber gloves, plastics.	100kg/h Capable of fully carbonizing animal bones for seafood markets and farms.	100-200kg/h (5,000kg/day) Large opening for backhoe loading; designed for long-exposed waste piles.	600kg/h (up to 13,200kg/day) Largest unit; capable of disposing of motorbike tires and large industrial waste.

ADVANCED FEATURES & SUSTAINABILITY

MULTI-CHAMBER INCINERATION



Heavy-duty models use multiple chambers with induced draft fans for total waste reduction.

ENERGY REPURPOSING

Industrial KGV 600 uses a 3600L/h water-cooling system generating hot water for pools.

MINIMAL OPERATIONAL COSTS

Operational expenses limited to air-blower electricity and rare kerosene use.